AutoDD: Learning to Select Drift Detectors for Evolving Data Streams

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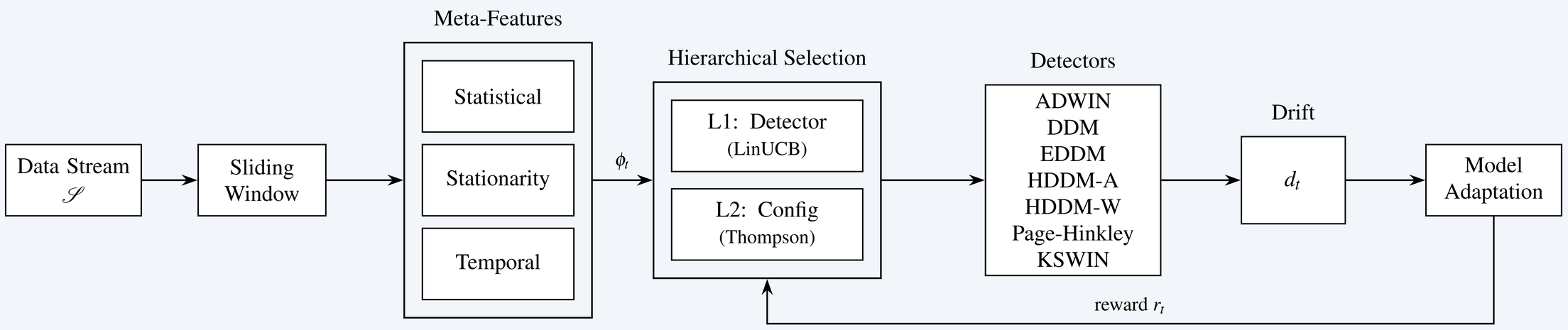
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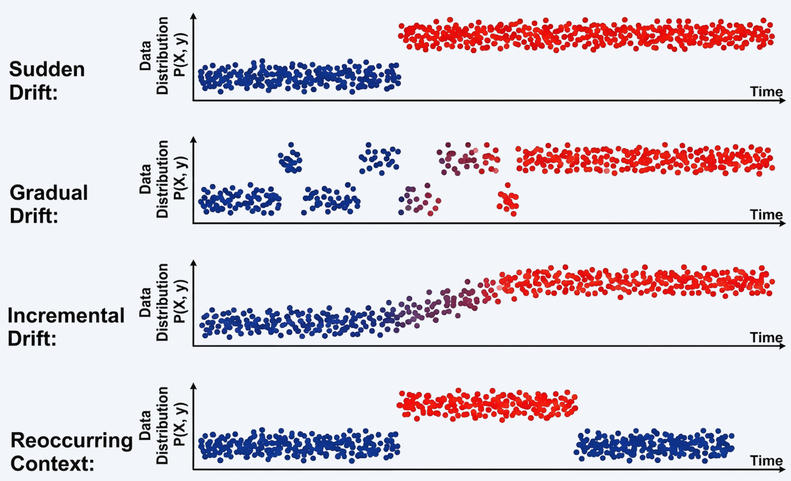
Code & Paper

Automated Drift Detector Framework



Overview of AutoDD. Meta-features extracted from a tumbling window form the context ϕ_t. LinUCB selects a detector, Thompson Sampling selects its configuration, and the resulting drift signal triggers adaptation. The reward r_t is then used to update both selection levels.

Introduction



Sudden, gradual, incremental, and recurring drift.

No single drift detector wins everywhere. Each one embeds an assumption about *how* drift arrives, so a detector tuned for abrupt change reacts slowly to gradual change, and the right choice shifts as the stream itself evolves.

Streaming AutoML automates the *learner*, but pins the *detector* to a fixed, hand-chosen setting, which leaves a gap in the automation loop.

Automating the choice online is hard because

- drift labels never arrive, so detector quality must be inferred from a **proxy reward** built from downstream model behaviour, and
- the joint space of detectors × configurations is far too large to search as one flat action set.

Problem Formulation

Given a portfolio $\mathcal{D} = \{D_1, \ldots, D_K\}$ where detector D_i has configuration space $\mathcal{H}_i$, AutoDD selects *jointly* and *online*

$$(D_t^*, \boldsymbol{\eta}_t^*) = \arg\max_{D_i \in \mathcal{D},\, \boldsymbol{\eta} \in \mathcal{H}_i} \mathbb{E}\big[\mathrm{Perf}\big(D_i(\boldsymbol{\eta}), \mathcal{S}_t\big) \mid \boldsymbol{\phi}_t\big]$$

with $\boldsymbol{\phi}_t$ the online stream context and $\mathrm{Perf}(\cdot)$ the downstream model performance.

Contributions

- **AutoDD**, the first framework to bring the drift detector itself into the automation loop, compatible with any model or AutoML pipeline.
- A **two-level hierarchical bandit** in which LinUCB over online meta-features chooses the detector and Thompson Sampling chooses its configuration.
- Proxy-regret analysis of both levels, plus an evaluation over **ten benchmarks** and two base learners where AutoDD takes the best average rank at competitive cost.

AutoDD Method

Context. 15 online meta-features are computed over a tumbling window and normalised to $[0, 1]$. They cover *statistical* properties (velocity, dimensionality, sparsity, noise, imbalance), *stationarity* (error volatility, KL divergence, autocorrelation, mean and variance shift), and *temporal* structure (trend, periodicity, change frequency and magnitude, recurrence).

Level 1, detector selection. Disjoint-arm LinUCB over $K = 7$ detectors

$$i_t = \arg\max_i \; \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i + \alpha_{\text{ucb}} \sqrt{\boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1} \boldsymbol{\phi}_t}$$

Level 2, configuration selection. Beta and Bernoulli Thompson Sampling over $C = 5$ configurations per detector.

Reward without labels. A bounded proxy $r_t \in [-1, 1]$ built from the windowed accuracy drop Δa_t, credited only when the change clears the significance threshold $\tau_{\text{sig}} = 0.05$, so noise-driven dips cannot reinforce spurious detections.

Theory

- **Level 1.** $R_1(T) = \tilde{O}(m\sqrt{KT})$ for LinUCB under bounded features, linear reward, and sub-Gaussian noise.
- **Level 2.** $R_2(T) = \sum_i O(\sqrt{C\, N_i(T) \log N_i(T)})$ for Thompson Sampling.

Both bound *idealised* sub-problems against the proxy objective, so they are surrogate analyses, not guarantees for the deployed system.

Results

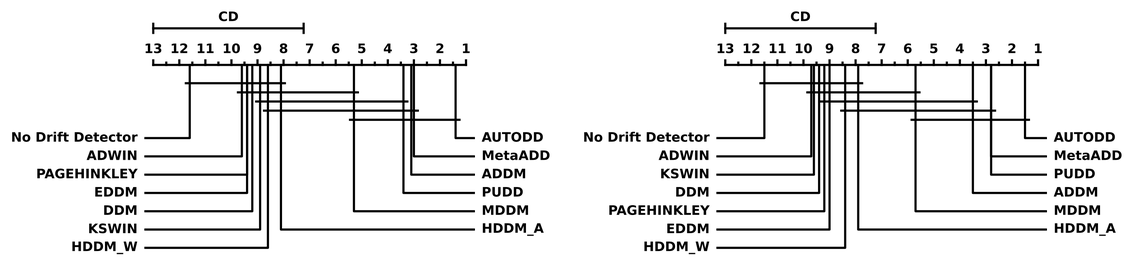
Mean over the 5 synthetic and 5 real-world benchmarks, against 12 baselines.

	GaussianNB		HAT	
Method	Syn. Acc	RW Acc	Syn. Acc	RW Acc
No Detector	69.5	66.6	72.1	69.4
ADWIN	74.3	71.5	78.0	74.8
HDDM-W	76.0	71.9	79.5	75.5
MDDM	76.4	74.0	79.7	77.0
ADDM	77.7	75.1	80.9	77.8
PUDD	78.1	74.6	81.6	77.8
Meta-ADD	80.9	74.8	**85.1**	77.6
AutoDD	**81.9**	**79.1**	84.8	**83.1**

Best macro-F1 on every synthetic stream (mean 0.86 / 0.90) and best real-world accuracy under both learners; Meta-ADD edges the HAT synthetic mean via RBF and RTG. Detection counts stay inside the consensus range of the compared methods.

Statistical Significance

Critical Difference diagrams from the Friedman–Nemenyi test ($\alpha = 0.05$). AutoDD attains the best average rank under both base learners.



GaussianNB **HAT**

Ablation Study

Variant	**Acc.**	**Δ**
Full AutoDD	**80.5%**	–
w/o Thompson (Level 2)	75.3%	−5.2
w/o Temporal	74.2%	−6.3
w/o Stationarity	73.5%	−7.0
w/o Statistical	71.2%	−9.3
w/o LinUCB (Level 1)	62.5%	−18.0

Matching the detector to the stream is the primary driver. Configuration search and all three meta-feature families add smaller, complementary gains, and AutoDD stays in the fast, low-memory quadrant against every comparator.

Conclusion and Future Work

The drift detector no longer has to be chosen by hand. A contextual bandit learns it online from a reward that needs no drift labels, and AutoDD attains the best average rank on ten benchmarks under both base learners at competitive computational cost.

Future work. Learning the window size w in an outer adaptive loop, replacing the accuracy-based proxy with a label-free reward, and granting baselines an equivalent tuning budget so that the benefit of selection is separated from the benefit of tuning.

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